

Paper-roll gate / design comparison

Baseline commit 732f592 | New experimental branch: iteration/paper-roll-click-magnet

Preferred trial: positive mechanical click joints, alternating diagonal webs, and removable magnetic paper pads. One PVC backing square and four corner braces remain the backbone.

Approved layout: four identical 2090.4 x 609.6 mm paper strips, 2700 mm outside and 1480.8 mm opening. Three gates plus spare paper per 100-foot roll. Four rigid borders travel with their paper attached.

Arrangement	Members per border	Main tradeoff
Ladder	12.64 m	Lowest material; depends on joint stiffness
Alternating diagonals	19.81 m	Triangles resist racking; preferred trial
Full crosshatch	26.98 m	36% more length than alternating; crossing nodes

Cost caution: at 150 mm maximum edge-pad spacing plus middle pads, the gate uses 168 clamp pairs / 336 magnets. At an assumed \$0.20 per magnet that is \$67.20, before plastic and PVC. At 300 mm spacing the trial count is 192 magnets. Holding and paper flutter need testing.

The heavy 12 mm diamond sample would use roughly 4.5 kg for all members in the alternating layout, before nodes. Lighter webs and a sparser structure need testing; a one- or two-spool finished gate is not established.

Print the interfaces before the gate

25 experimental STL variants, not 25 required production part types. No printer job has been sent.

Start with plate 05: seven pieces test the D-shaped magnetic holder and complete PVC dock. Add a paper scrap, two 6 x 2 mm magnets and actual 33.4 mm OD PVC. The dock needs four M3x25 and two M3x16 bolts, six nuts and twelve washers.

Plate / experiment	PETG	Slicer estimate
05_D_MAGNET_AND_DOCK	33.2 g	1 h 53 min
01_CLICK_AND_MAGNET_JOINTS	23.9 g	1 h 05 min
03_STACK_SEPARATION_TESTS	33.3 g	1 h 34 min
DIAGONAL_8x220	104.3 g	4 h 19 min
06_OVERNIGHT_26_RAILS	301.4 g	11 h 54 min

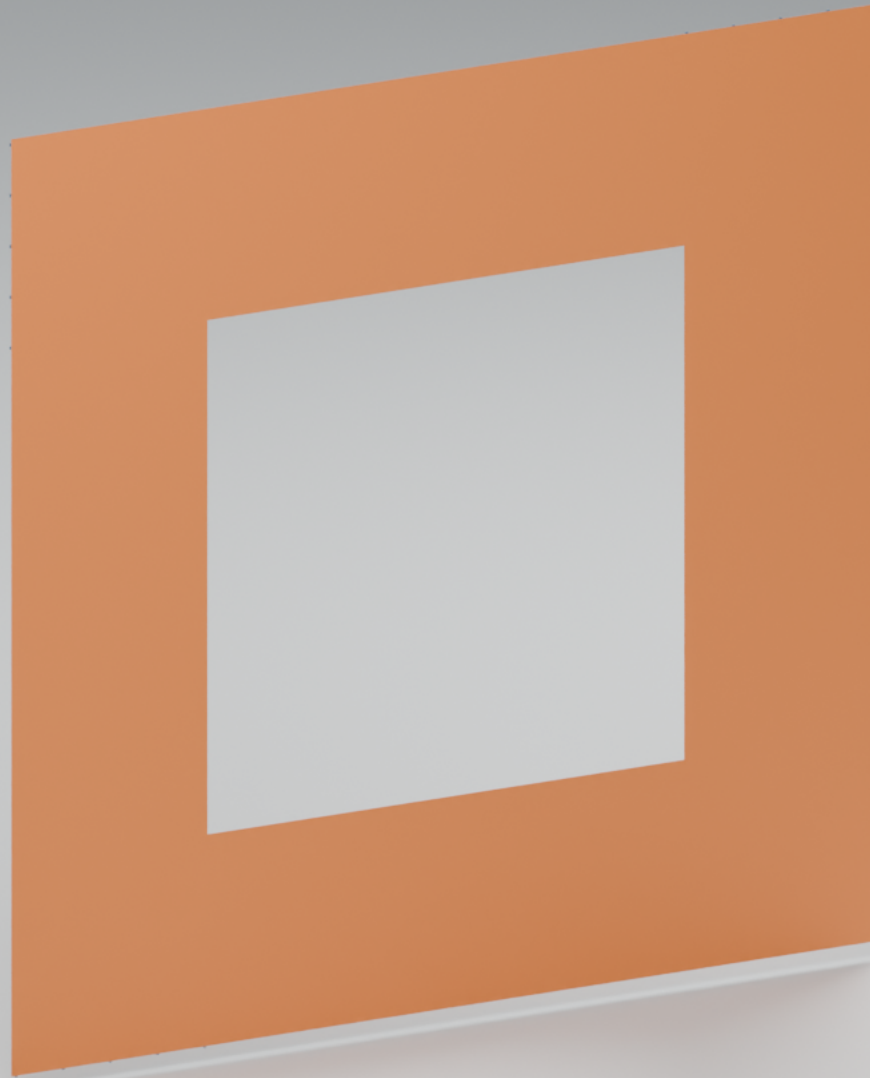
Diagonal batch: two stacks of thirteen 198 mm rails fit 169.7 x 169.7 x 161.8 mm. They are plain manufacturing samples, without final rail-end joints. Test short stacks before the overnight plate.

Stack findings: 0.4 mm necks vanished in the standard slicer profile. Current continuous trials use 0.6 and 0.8 mm; every neck layer has extrusion. End tabs removed an initial cantilever warning. All current slices have supports off and no warnings; physical bridging and PETG separation remain untested.

Verified digitally: 25 manifold masters, 31 actual A1 Mini/PETG slices and 20 assembled-pair interference checks. The gate overview is schematic; production corner/diagonal nodes and brace interfaces remain to be integrated after measured fit tests.

Paper-roll gate / three joint strategies

A: clicks + snap battens. B: keyed magnetic joints + magnetic pads. C: click frame + magnetic pads (recommended experiment).

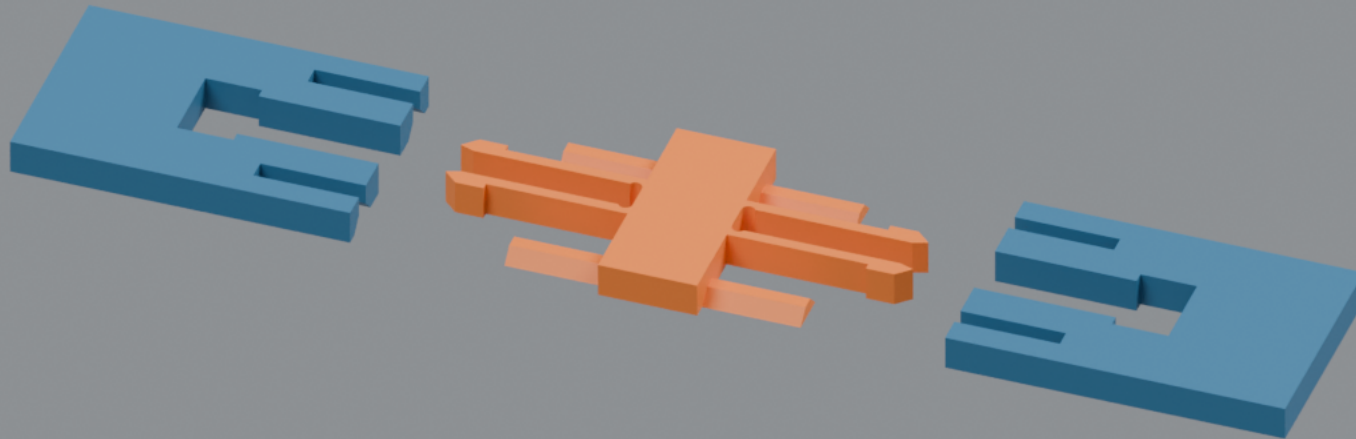


2700 mm outside
1480.8 mm opening
609.6 mm paper width

Four strips: 2090.4 mm each
Three complete gates per 100 ft roll
Zero zip ties in proposed joinery

Replaceable click key / exploded

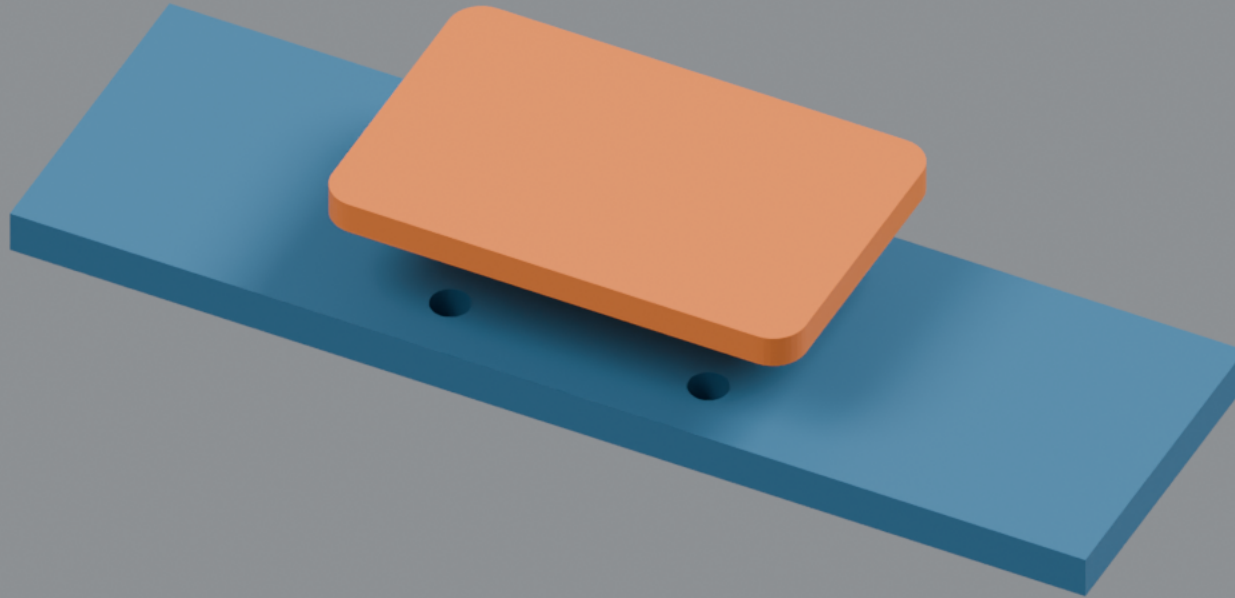
Rigid dovetail guides take alignment and lift loads. Two in-plane fork latches prevent axial pullout; pinch both prongs to release.



Test 0.25 and 0.40 mm socket clearances. Original prototype geometry, not a Kato-compatible part. Verify both guide fit and complete hook engagement.

Magnets retain; pins locate

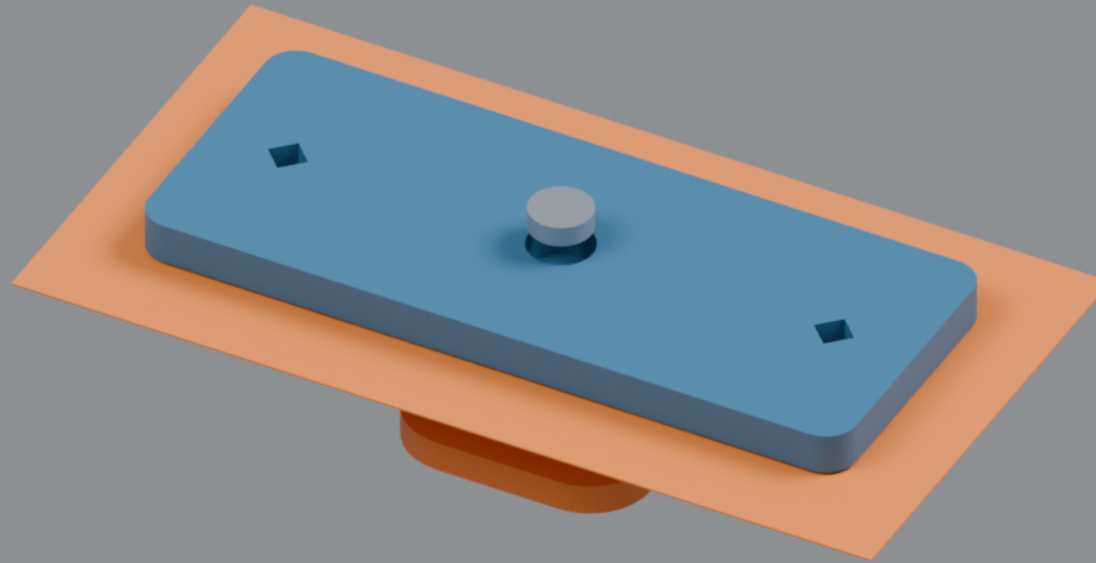
Two identical rail-end samples and one keyed cap. Four 6 x 2 mm magnets per completed joint; separate by lifting the cap.



This joint can release under peel. It is a comparison coupon, not an approved structural gate connector. Magnetic pull is not a shear rating.

Magnet / thin frame / paper / magnet

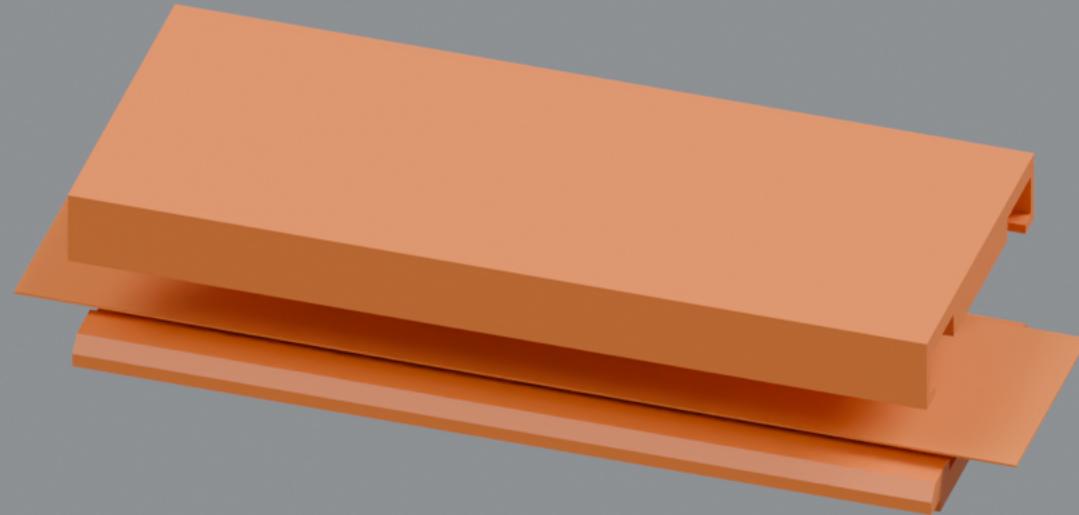
Compare 0.4, 0.8 and 1.2 mm frame skins. The broad front pad adds a 0.4 mm skin to spread pressure and protect the paper.



Rear-load pockets. Check polarity and pocket size before adhesive retention. Test normal pull, edge peel and sliding with the actual roll paper.

Snap-over paper batten

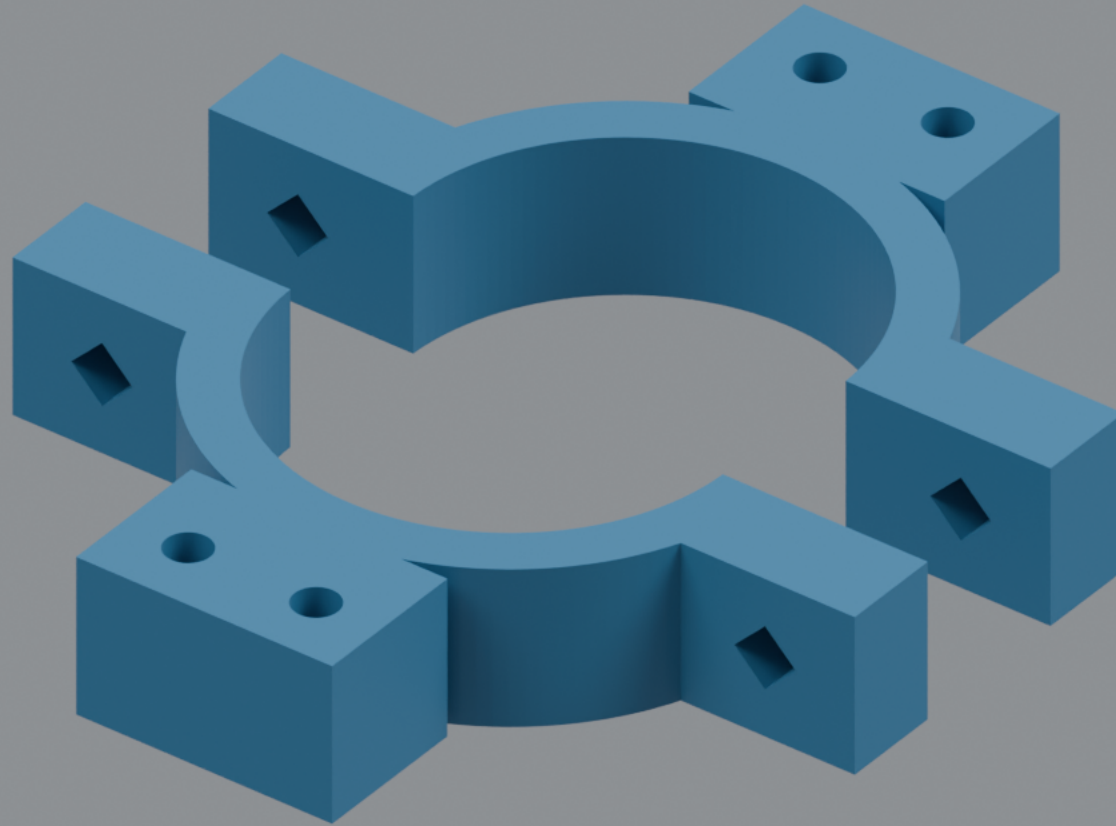
A magnet-free comparison: push the cap over the paper and base. Print both end-up so the spring section lies within print layers.



Two clearance trials are supplied. Assess paper slip, creasing and release; clamping thin paper without tearing it is not yet demonstrated.

PVC collar without zip ties

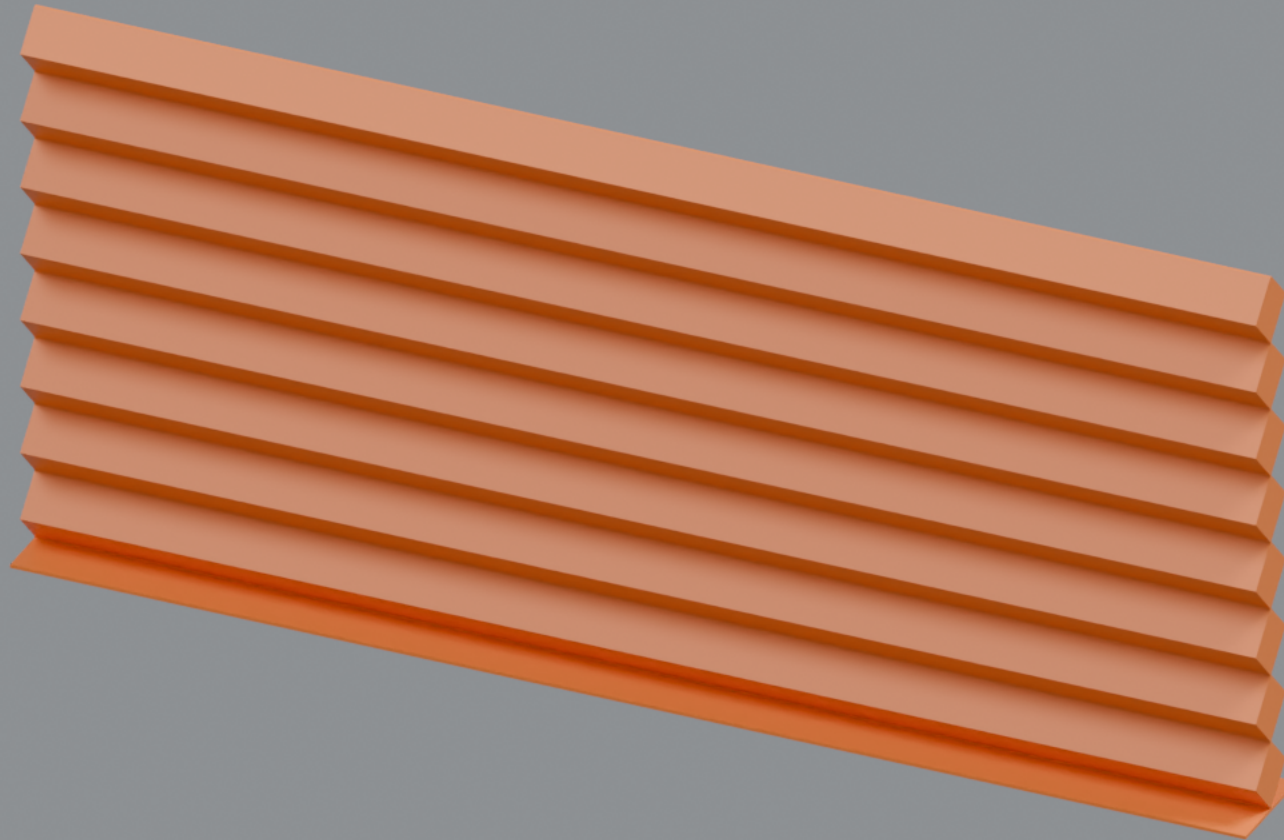
Two identical halves; two M3 bolts, nuts and washers provide clamp preload. This workshop attachment can stay on the pipe.



Use the mating dock in scene 13 for the 33.4 mm collar. A production angled brace node remains to be designed after collar grip tests.

Eight long rails in one vertical stack

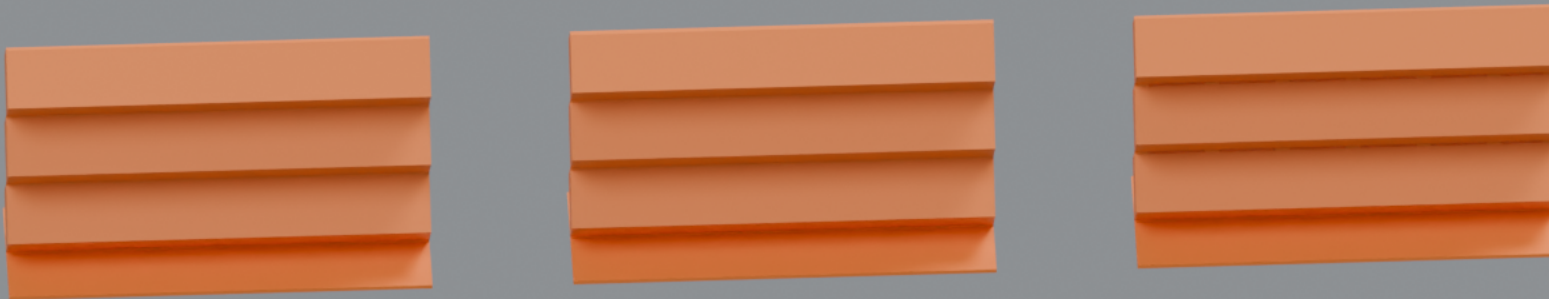
220 mm rails. Diamond sections expand at 45 degrees above 0.6 mm sacrificial necks. Starter foot is removed with the separators.



Exact print-strategy coupon, shown before bed rotation. Same-material PETG necks fuse: separation force and damage must be tested, not assumed.

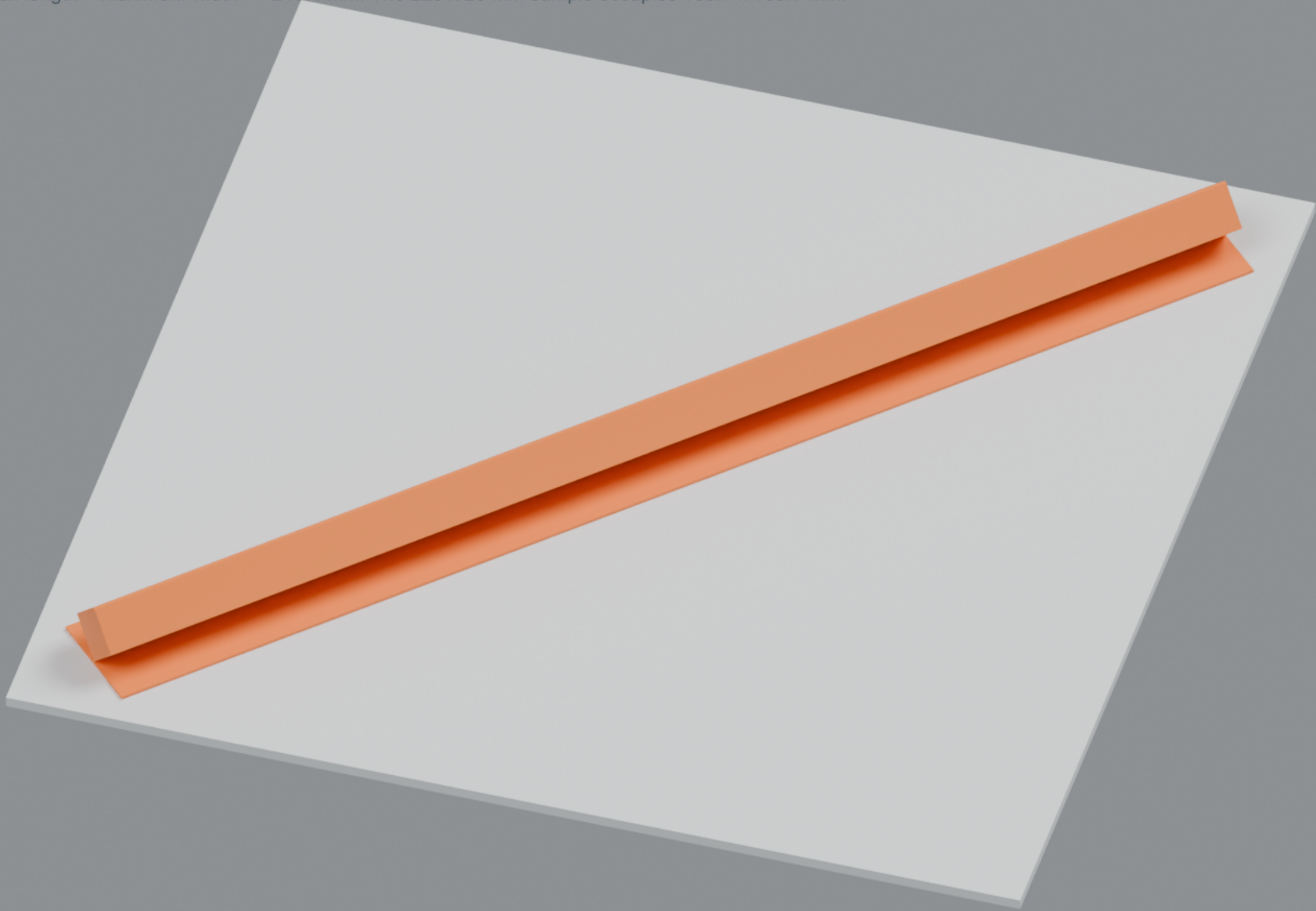
Score, cut or break: compare first

Three short stacks: 0.6 mm continuous neck, 0.8 mm continuous neck, and 2 mm tabs with 8 mm bridges between them.



The diagonal has a width budget

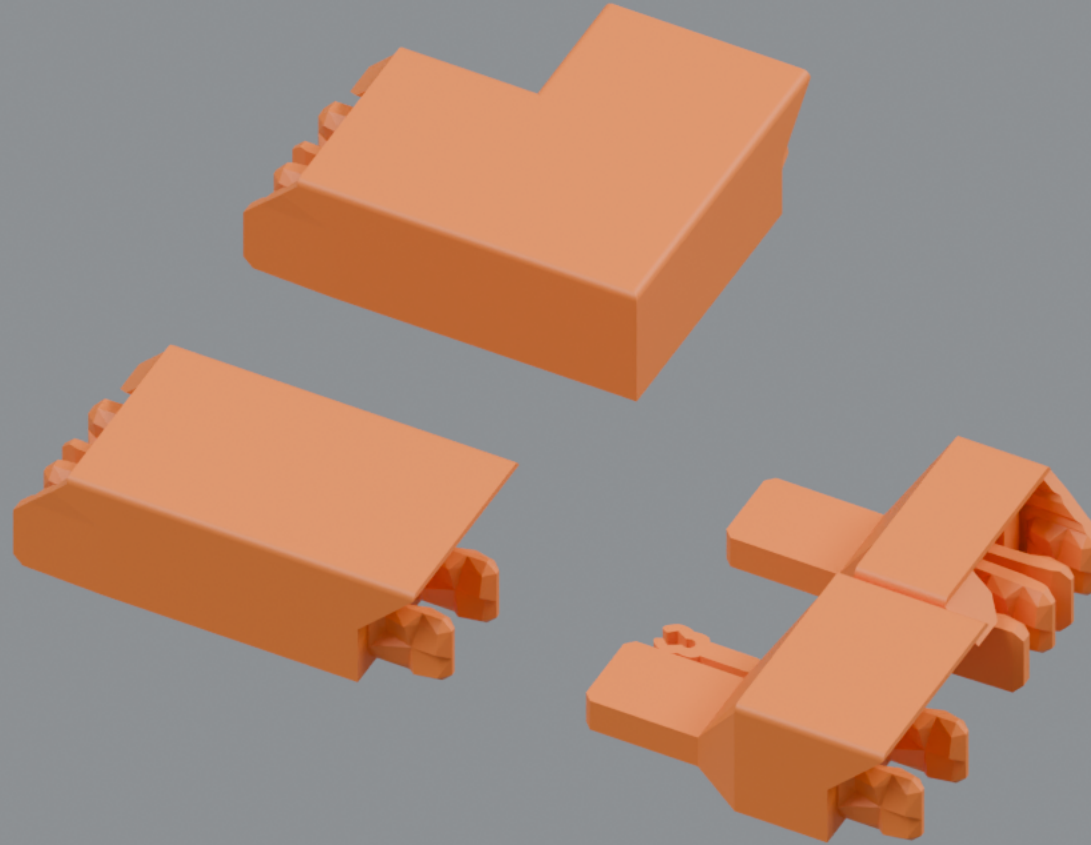
With 5 mm edge margin: rail length + maximum width ≤ 240.4 mm. The 220 x 20 mm sample occupies 169.7 x 169.7 mm.



All tongues, hooks, starter feet and brims count. The nominal 254.6 mm diagonal is a zero-width line; stack height must independently fit below 180 mm.

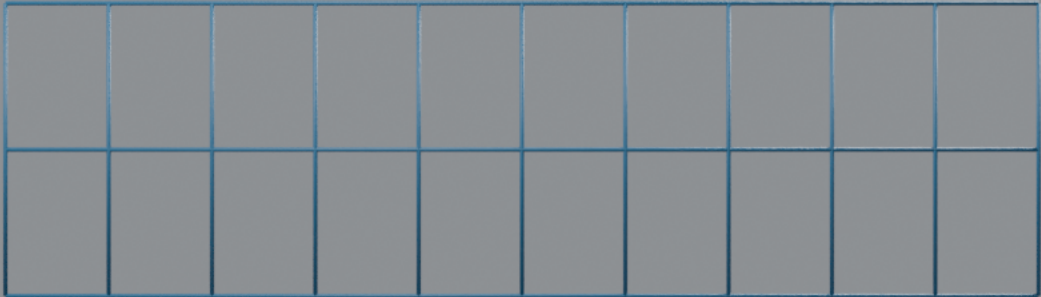
Your picture-frame reference

Repeated snap teeth, modular lengths and a calibration coupon. Adapt the concept with accessible release and rigid alignment guides.

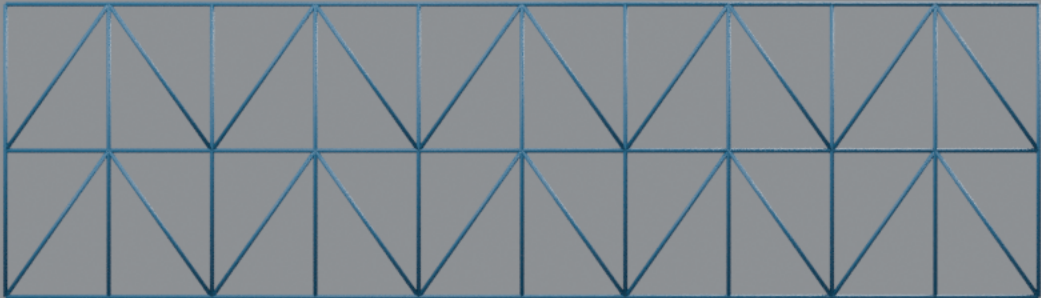


Join the two edges with triangles

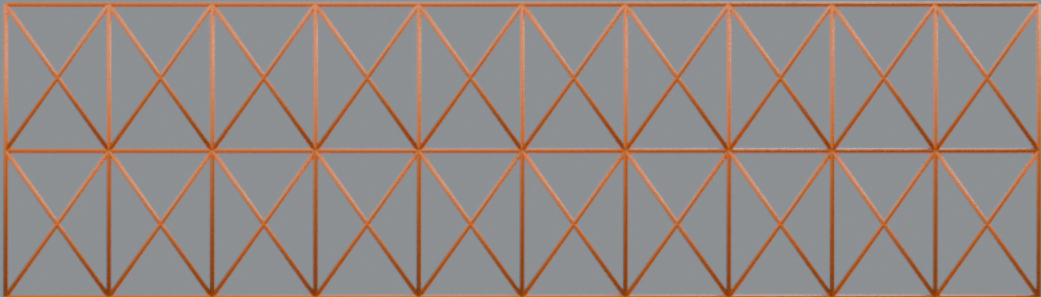
Ladder: simplest, relies on stiff joints. Alternating diagonals: preferred trial. Crosshatch: redundancy at higher material and joint count.



LADDER / 12.64 m of members per border



ALTERNATING / 19.81 m of members per border

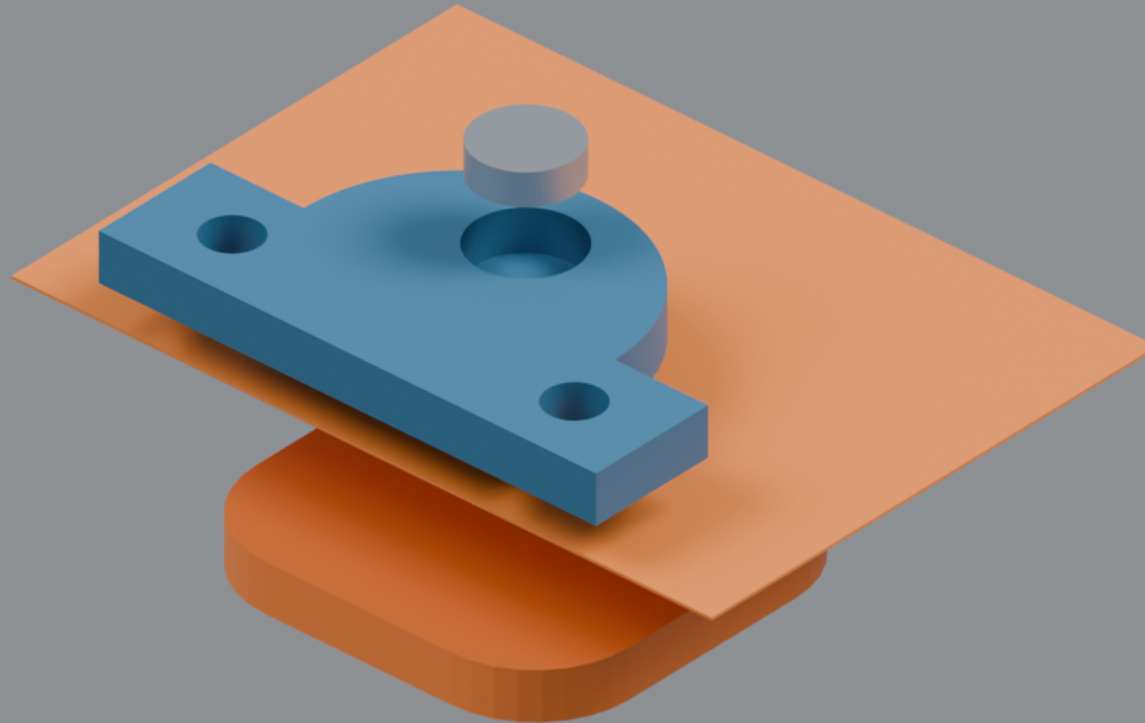


CROSSHATCH / 26.98 m of members per border

All three include a middle support rail. Geometry comparison only: identical sections assumed; joint compliance, bending and wind performance need physical tests.

Half-cylinder magnet holders

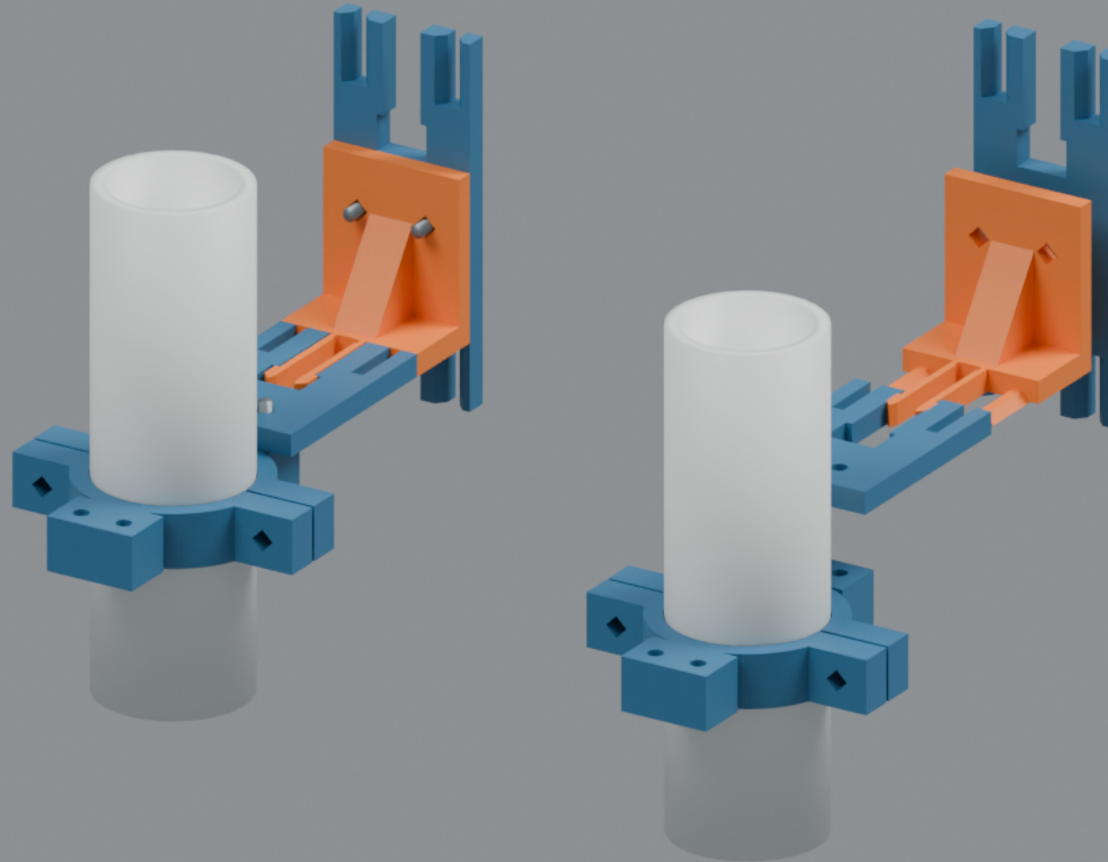
A D-shaped tab projects inside the rail. Rear magnet sits behind a 0.4 mm window; a removable front pad sandwiches the paper.



Exploded. Same lug works along an edge; rotate it at corners. Flat flange is a bolt-on test interface, to become an integral rail/node feature after fit tests.

Frame node clicks into a PVC dock

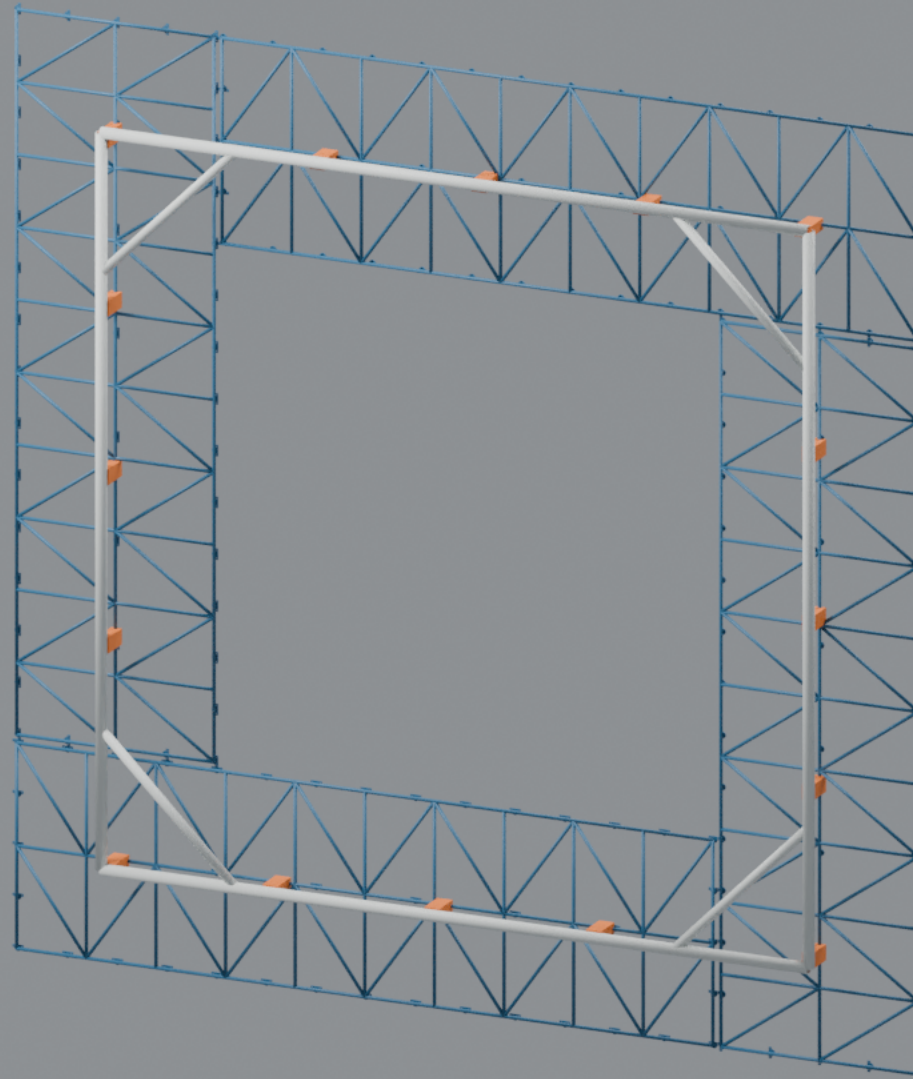
Left: assembled. Right: exploded. Bolt the receiver to the split collar; slide in the frame tongue until both catches engage.



Collar stays on PVC; frame tongue stays on the border. Pinch both fork tips at the open rear throat to withdraw. Positive guides carry load; latch retains.

One PVC back / four triangulated borders

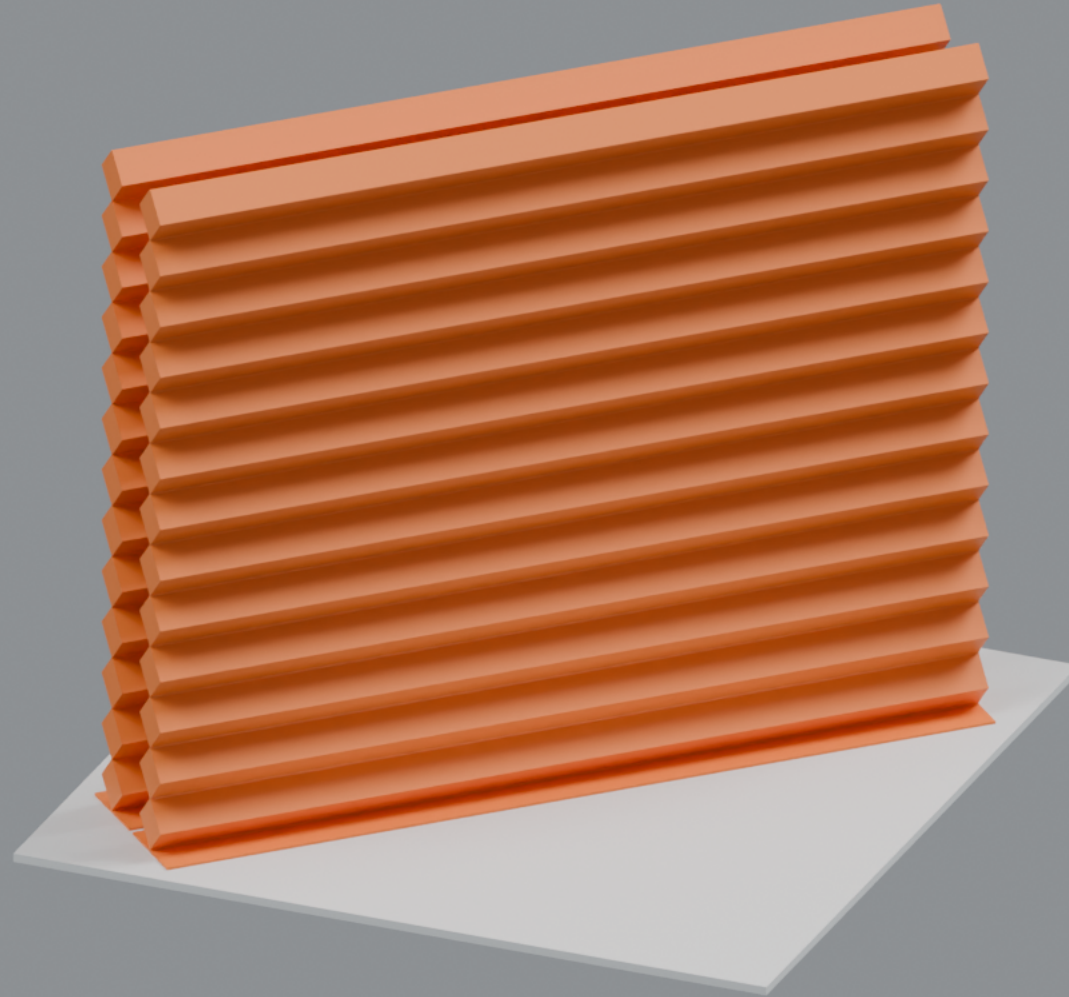
Paper removed. Edge rails and diagonal webs feed the center rail; four spaced docks per border transfer load to the PVC backing.



Dock positions and rail/node geometry are conceptual. PVC remains the primary backbone. Add staked or ballasted feet for the surface; the paper is not structural bracing.

More rails per overnight batch

Two diagonal stacks, 13 rails each: 26 x 198 mm. A 2 mm gap separates the starter feet; overall footprint 169.7 mm square.



161.8 mm tall. Shorter rails trade more eventual joints for 1.8x the rail length per batch versus thirteen 220 mm rails. Test short neck coupons first.